



University of Asia Pacific

Department of Computer Science & Engineering (CSE)

Course Title: Database Systems Lab

Course Code: CSE 212

District-Based Database Management

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Project Description: District-Based Database Management System

The District-Based Database Management System (DBMS) is a comprehensive data management solution designed to handle diverse and interrelated datasets at the division and district levels. It provides a robust framework for organizing and analyzing information such as demographics, transport, culture, education, and healthcare across various regions. The project is particularly focused on addressing the needs of administrative and governance bodies, tourism boards, and developers looking to create applications powered by regional data.

Key Components:

1. Administrative Data:

- Division and district-level information, including population, area, land types, and headquarters.

2. Cultural Data:

- Famous places, cultural events, local languages, and notable individuals from each district.

3. Transport Data:

- Detailed transport routes connecting districts, categorized by transport type, with distance and fare information.

4. Education and Health:

- Comprehensive data on educational institutions and healthcare facilities, including their types, capacities, and rankings.

5. Economic and Demographic Data:

- Insights into industries, employment, and demographic trends across divisions and districts.

Users:

The District-Based Management System is designed to streamline the management and allocation of resources, services, and information at the district level. The primary users of this system include:

1. District Administrators

- Responsible for overseeing the district's operations, ensuring efficient service delivery, and managing district-level resources and activities.

2. Department Heads

- Leaders of various district departments (e.g., healthcare, education, infrastructure) who use the system to monitor progress, track budgets, and allocate resources.

3. Field Officers

- Personnel working in the field who collect data, provide on-the-ground insights and ensure that resources are effectively distributed across the district.

4. Local Government Officials

- Elected representatives and officers who utilize the system to stay updated on district affairs, policy enforcement, and constituent services.

5. Public Service Agencies

- Government agencies and organizations responsible for providing essential public services, including health, education, and emergency management, using the system to manage operations and track outcomes.

Entities & Attributes:

1. Division

- Attributes:
 - Division_ID (INT)
 - Name (VARCHAR)
 - Size_KM2 (DECIMAL)
 - Population (INT)
 - EstablishedYear (INT)
 - Landtype (VARCHAR)

2. District

- Attributes:
 - District_Name (VARCHAR)
 - Division_ID (INT, Foreign Key referencing Division)
 - Area (DECIMAL)
 - Population (INT)
 - Education_Percentage (DECIMAL)
 - Postal_Code (VARCHAR)

3. Transport_Type

- Attributes:
 - Transport_Type_ID (INT)
 - Transport_Type_Name (VARCHAR)

4. Transport_Route

- Attributes:
 - Route_ID (INT)
 - Start_Point (VARCHAR, Foreign Key referencing District)
 - End_Point (VARCHAR, Foreign Key referencing District)
 - Distance (DECIMAL)
 - Transport_Type_ID (INT, Foreign Key referencing Transport_Type)
 - Fare (DECIMAL)

5. Education_Type

- Attributes:
 - Education_Type_ID (INT)
 - Type_Name (VARCHAR)
 - Total_count (INT)
 - District_Name (VARCHAR, Foreign Key referencing District)

6. District_Famous_Person

- Attributes:
 - Person_ID (INT)
 - Full_Name (VARCHAR)
 - Profession (TEXT)
 - Notable_Work (TEXT)
 - District_Name (VARCHAR, Foreign Key referencing District)

7. Famous_Food

- Attributes:
 - Food_ID (INT)
 - Food_Name (VARCHAR)
 - Type (VARCHAR)
 - Popularity_Score (INT)
 - Description (TEXT)
 - Food_Price_Range (VARCHAR)
 - District_Name (VARCHAR, Foreign Key referencing District)

8. LocalLanguages

- Attributes:
 - District_Name (VARCHAR, Primary Key, Foreign Key referencing District)
 - Normal_Language (VARCHAR)
 - Local_Language (VARCHAR)
 - Understandability_Rate (DECIMAL)

9. District_Health_Facilities

- Attributes:
 - Facility_ID (INT)

- Facility_Name (VARCHAR)
- Address (VARCHAR)
- Contact (VARCHAR)
- Patient_Capacity (INT)
- District_Name (VARCHAR, Foreign Key referencing District)

10. Cultural_Event

- Attributes:
 - Event_ID (INT)
 - Event_Name (VARCHAR)
 - Event_Date (DATE)
 - Description (TEXT)
 - Event_Type (VARCHAR)
 - Location (VARCHAR)
 - District_Name (VARCHAR, Foreign Key referencing District)

11. FamousPlaces

- Attributes:
 - Place_ID (INT)
 - Place_Name (VARCHAR)
 - Description (TEXT)
 - Visitor_Per_Year (INT)
 - Entry_Fee (DECIMAL)
 - Opening_Time (TIME)
 - District_Name (VARCHAR, Foreign Key referencing District)

Relationships:

1-to-1 Relationships

1. District ↔ LocalLanguages

- A **District** has exactly one **LocalLanguages** entry.
- Keys: District_Name (PK in both District and LocalLanguages).

1-to-Many Relationships (1:M)

1. Division ↔ District

- A **Division** contains many **Districts**.
- Keys: Division_ID (PK in Division, FK in District).

2. District ↔ Education_Type

- A **District** can have multiple **Education_Types**.
- Keys: District_Name (PK in District, FK in Education_Type).

3. Education_Type ↔ Educational_Institution

- An **Education_Type** can have multiple **Educational_Institutions**.
- Keys: Education_Type_ID (PK in Education_Type, FK in Educational_Institution).

4. District ↔ District_Famous_Person

- A **District** can have multiple **Famous_Persons**.
- Keys: District_Name (PK in District, FK in District_Famous_Person).

5. District ↔ Famous_Food

- A **District** can have multiple **Famous_Foods**.
- Keys: District_Name (PK in District, FK in Famous_Food).

6. Transport_Type ↔ Transport_Route

- A **Transport_Type** can be associated with multiple **Transport_Routes**.
- Keys: Transport_Type_ID (PK in Transport_Type, FK in Transport_Route).

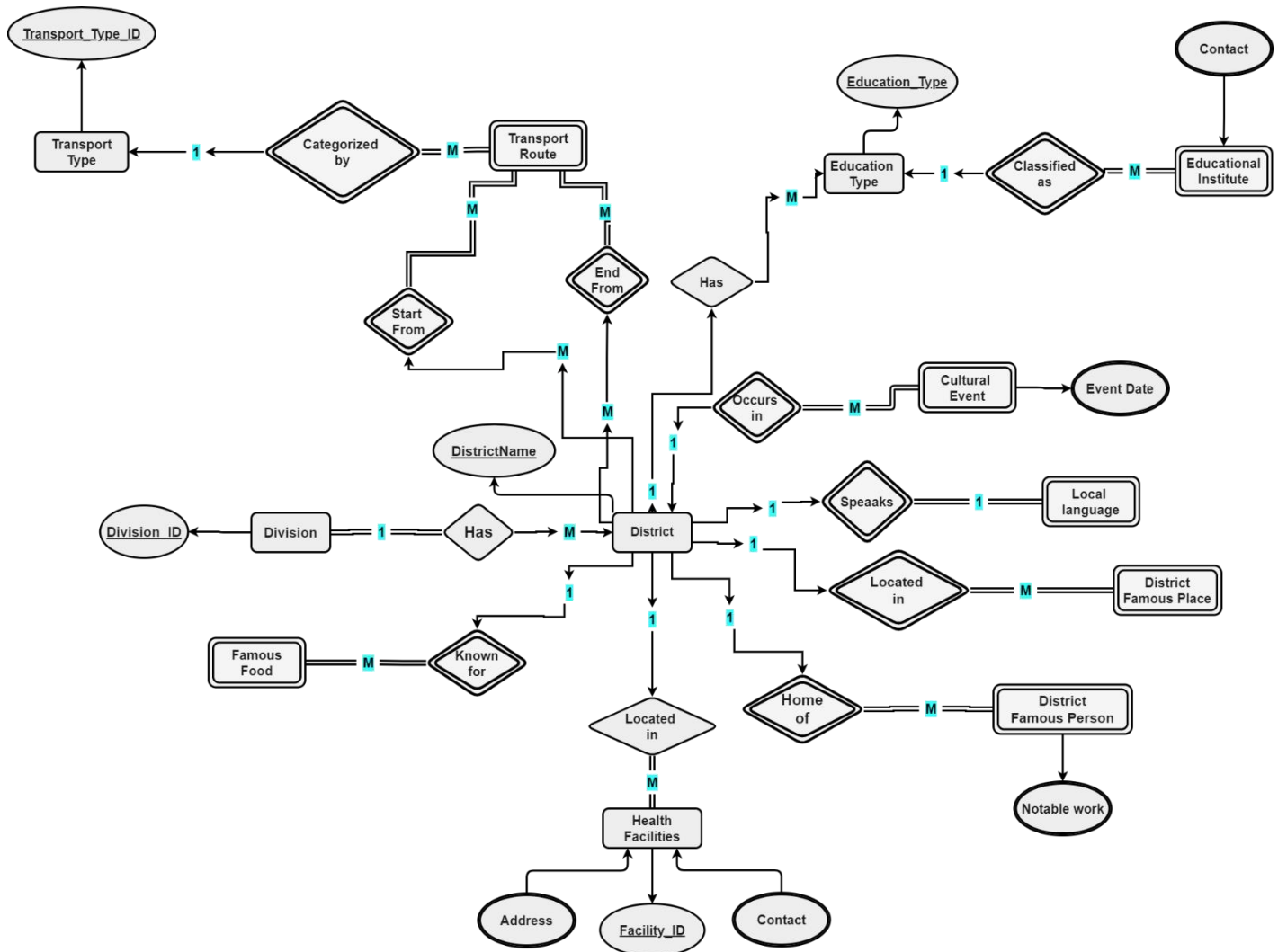
Many-to-Many Relationships (M:M)

Represented via bridge tables or implied associations:

1. District ↔ Transport_Route

- A **District** can be the start or end point of multiple **Transport_Routes**, and a **Transport_Route** can span multiple **Districts**.
- Keys: Start_Point, End_Point (FK in Transport_Route, PK in District).

E-R Diagram



Schema Diagram



Query Questions & Tables:

Level 1:

Q1. Find the names of all districts where the education percentage is greater than 75%.

Query:

```
SELECT District_Name, Education_Percentage
FROM District
WHERE Education_Percentage > 75;
```

District_Name	Education_Percentage
Bagerhat	75.50
Cumilla	84.30
Dhaka	88.00
Mongla	75.50
Narshingdi	81.10
Netrokona	81.50
Patuakhali	75.50
Sylhet	75.33
Tangail	75.80

Q2. What are the names and descriptions of all famous foods from the district "Chattogram"?

Query:

```
SELECT Food_Name, Description
FROM Famous_Food
WHERE District_Name = 'Chattogram';
```

Food_Name	Description
Chittagong Mezban Beef	Spicy beef dish served at traditional Mezban feasts.

Q3. List all the districts with a population greater than 5,000,000.

Query:

```
SELECT District_Name, Population
FROM District
WHERE Population > 5000000;
```

District_Name	Population
Chattogram	7614253
Dhaka	8906039
Mymensingh	5116654

Level 2:

Q1. Find all districts with famous persons and their professions.

Query:

```
SELECT d.District_Name, dfp.FULL_NAME, dfp.PROFESSION
FROM District d
INNER JOIN District_Famous_Person dfp ON d.District_Name = dfp.DISTRICT_NAME;
```

District_Name	FULL_NAME	PROFESSION
Chattogram	Muhammad Yunus	Economist
Cox's Bazar	Mohammad Abdul Momen	civil servant
Bandarban	Bir Bikram U. K. Ching	Freedom Fighter
Khagrachari	Priti Kumar Mitra	Historian
Rangamati	Basanti Chakma	Activist
Chandpur	Shaikh Siraj	Agriculturist & Journalist
Noakhali	Obaidul Quader	Politician
Cumilla	Nawab Faizunnesa Choudhurani	only female Nawab in South Asia
Gopalganj	Sheikh Hasina	Former President
Gazipur	Sufi Mohammad Sharif	Philosopher
Sylhet	Hason Raja	Poet & Philosopher
Dinajpur	Nural Hoda Choudhury	politician

Q2. Find the average fare of all transport routes and list routes with fares above the average.

Query:

```
SELECT tr.Route_ID, tr.Start_Point, tr.End_Point, tr.Fare
FROM Transport_Route tr
WHERE tr.Fare > (SELECT AVG(Fare) FROM Transport_Route);
```

Route_ID	Start_Point	End_Point	Fare
11	Dhaka	Sylhet	2500
12	Dhaka	Chattogram	3500
13	Dhaka	Khulna	4500

Q3. List districts grouped by their division, showing the average education percentage for each division.

Query:

```
SELECT d.Division_ID, AVG(d.Education_Percentage) AS Avg_Education
FROM District d
GROUP BY d.Division_ID
ORDER BY Avg_Education ASC;
```

Division_ID	Avg_Education
8	45.300000
7	65.006667
2	66.730000
5	68.870000
4	70.775000
6	71.832500
3	73.120000
1	76.442857

Level 3:

Q1: Find districts grouped by their division and list only divisions where the average population exceeds 3,000,000.

Query:

```
SELECT d.Division_ID, AVG(d.Population) AS Avg_Population
FROM District d
GROUP BY d.Division_ID
HAVING AVG(d.Population) > 3000000
ORDER BY Avg_Population DESC;
```

Division_ID	Avg_Population
1	3987952.1429
2	3673772.6667

Q2: List all districts grouped by division, showing the total number of famous foods for each division where the total famous foods exceed 3.

Query:

```
SELECT d.Division_ID, COUNT(ff.Food_ID) AS Total_Foods
FROM District d
INNER JOIN Famous_Food ff ON d.District_Name = ff.District_Name
GROUP BY d.Division_ID
HAVING COUNT(ff.Food_ID) > 3
ORDER BY Total_Foods DESC;
```

Division_ID	Total_Foods
1	5

Q3: Count the number of transport routes for each transport type.**Query:**

```
SELECT tt.Transport_Type_Name, COUNT(tr.Route_ID) AS Route_Count
FROM Transport_Type tt
INNER JOIN Transport_Route tr ON tt.Transport_Type_ID = tr.Transport_Type_ID
GROUP BY tt.Transport_Type_Name;
```

Transport_Type_Name	Route_Count
Bus	4
Train	3
Boat	3
Airplane	3

CEP Mapping

Mapping Knowledge Profiles (K's), Course Outcomes (COs), and Program Outcomes (POs):

K's	Attributes	How K's are Addressed Through the Project	COs	POs
K2	Mathematics	Statistical and analytical knowledge is applied for calculating population percentages, visitor statistics, and resource allocations across districts.	CO2, CO3, CO4	PO1, PO3, PO5, PO6, PO7, PO8
K3	Engineering Fundamentals	Understanding database fundamentals is crucial for designing efficient relationships between entities like Division, District, and Famous Places.	CO2, CO3, CO4	PO1
K5	Engineering Design	Designing ER and schema diagrams allows efficient organization of data such as transport routes, famous places, and health facilities.	CO3, CO4	PO3, PO5
K6	Engineering Practice	SQL is used for database creation, management, and advanced queries to ensure smooth implementation of the system, maintaining data accuracy.	CO1, CO2, CO5	PO5

Complex Engineering Problems (P's) Addressed Through Our Project:

P's	Attributes	How P's are Addressed Through the Project	COs	POs
P1	Problem Identification	The project identifies the need for an organized system to manage district-level data such as population, resources, transport, and health services.	CO1, CO2, CO3	PO1, PO2
P2	Problem Complexity	Handling interrelated data among districts, divisions, and services requires solving abstract and structured problems in database design and queries.	CO3, CO4, CO5	PO1, PO2, PO4
P3	Problem-Solving Approach	Developing an efficient and scalable database involves a logical approach to solving problems using ER modeling, schema design, and optimization.	CO3, CO4	PO3, PO5
P5	Use of Engineering Tools	SQL and tools like MySQL Workbench are used to create, manage, and query the database effectively, ensuring the practical application of theoretical concepts.	CO1, CO5	PO5, PO11

Complex Engineering Activities (A's) Addressed Through Our Project:

A's	Attributes	How A's are Addressed Through Our Project	COs	POs
A1	Range of Resources	The project utilizes diverse resources, including district data (population, resources, transport, and health), technologies like SQL tools, and people such as data analysts and engineers. Efficient management of these resources ensures the project's success.	CO8	PO11
A4	Consequences for Society and Environment	The project has positive consequences for society by enhancing district-level governance, improving public services, and supporting decision-making for resource allocation. It promotes better healthcare, transport, and education services.	CO6	PO6, PO7, PO8
A5	Familiarity	The project requires expertise in database design, implementation, and optimization, as well as familiarity with software tools like MySQL Workbench and SQL queries. Ensuring team familiarity with these activities is key to project success.	CO9	PO1, PO5

~~ The End, Thank you! ~~